

Limits and Continuity Exercises

A. True or false? If true, explain why. If false, give a counter-example.

1. If $\lim_{x \rightarrow a} f(x)$ does not exist, then f is undefined at the point $x = a$.
2. If a function is not defined at $x = a$, then $\lim_{x \rightarrow a} f(x)$ does not exist.
3. If f and g are continuous on their domains which contain a , then $\lim_{x \rightarrow a} (f(x) + g(x)) = f(a) + g(a)$.
4. If $\lim_{x \rightarrow a} f(x)$ exists, then f is continuous at a .

B. Evaluate the following limits using the limit laws. Write which law you are using in each step.

1. $\lim_{x \rightarrow 4} x^2 + 3x - 1$
2. $\lim_{x \rightarrow -2} \sqrt[3]{x^4 + 1}$
3. $\lim_{x \rightarrow 0} (\sqrt{x} + 1)^{100}$
4. $\lim_{x \rightarrow 2} \frac{x - 3}{x + 4}$

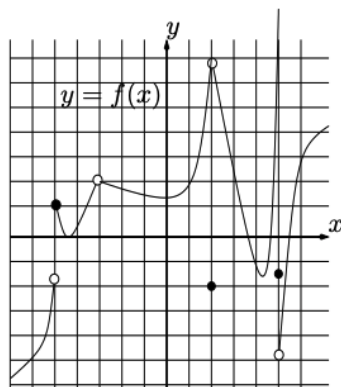
C. Evaluate the following limits (or say that the limit DNE):

1. $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x + 3}$
2. $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$
3. $\lim_{x \rightarrow \pi/2} \frac{\cot(x)}{\cos(x)}$
4. $\lim_{x \rightarrow 6} \frac{10}{x^2 - 36}$
5. $\lim_{x \rightarrow \infty} \tan(x)$
6. $\lim_{x \rightarrow \pi/2^+} \tan(x)$
7. $\lim_{x \rightarrow \infty} \arctan(x)$
8. $\lim_{x \rightarrow \infty} \frac{x^3 + 3x^2 + 4}{1 - x^2}$
9. $\lim_{x \rightarrow \infty} \frac{\cos(x)}{x^2}$
10. $\lim_{x \rightarrow \infty} \frac{4x^4 + 3x^3}{7x^4 + x}$
11. $\lim_{x \rightarrow \infty} \frac{10000x^3 - x^2}{8x^4 + 2x + 1}$
12. $\lim_{x \rightarrow 1^+} \frac{x^2 + x + 1}{x^2 - 1}$
13. $\lim_{x \rightarrow 0} \frac{(\cos^2(x) - 1)(x + 3)}{x}$
14. $\lim_{x \rightarrow 5} x^3 + e^x \sin(x)$
15. $\lim_{x \rightarrow 5} \frac{6 \sin(x - 5)}{x - 5}$
16. $\lim_{x \rightarrow 0} \frac{\sin(x^2)}{x}$
17. $\lim_{x \rightarrow \infty} \frac{\sqrt{3x^4 - x^2} + x^2}{4x^2 + 1}$
18. $\lim_{x \rightarrow 0^+} e^{-1/\ln(x)}$
19. $\lim_{x \rightarrow \infty} \left(1 + \frac{4}{3x}\right)^x$
20. $\lim_{x \rightarrow 0} x^2 \cos\left(\frac{1}{x}\right)$

D. For each function f , find a value of c so that f is continuous on $\mathbb{R}$.

1. $f(x) = \begin{cases} 2x & x \leq c \\ x^2 + 1 & x > c. \end{cases}$
2. $f(x) = \begin{cases} 2x + c & x < 2 \\ x^2 + cx + 1 & x \geq 2. \end{cases}$

E. Answer the following questions based on the graph (each box has width 1).



1. At what points a does $\lim_{x \rightarrow a} f(x) = L$ but $L \neq f(a)$?
2. At which points is f continuous?
3. At which points is f not continuous?
4. Does $\lim_{x \rightarrow 2^-} f(x)$ exist? If it does, what is its value?
5. Does $\lim_{x \rightarrow 2^+} f(x)$ exist? If it does, what is its value?
6. Does $\lim_{x \rightarrow 2} f(x)$ exist? If it does, what is its value?
7. What is $f(2)$?

F. Answer the following questions based on the function f defined below.

$$f(t) = \begin{cases} 1+t & t < 0 \\ t^2 + 1 & 0 \leq t < 1 \\ 3 & t = 1 \\ t+4 & t > 1 \end{cases}$$

1. What is $\lim_{t \rightarrow 0} f(t)$?
2. What is $\lim_{t \rightarrow 0^+} f(t)$?
3. What is $\lim_{t \rightarrow 0^-} f(t)$?
4. Where is f continuous?

G. Describe the points of discontinuity of the following functions.

$$1. f(x) = \begin{cases} \frac{\sin(3x^2)}{x-1} & 0 \leq x \\ \frac{(x^2+1)(x-2)}{x^2-4} & x < 0 \end{cases}$$

$$2. f(x) = \begin{cases} \frac{x-5}{x^2-7x+10} & 2 < x \\ \frac{1}{1-2^{(x-2)}} & -1 \leq x < 2 \\ \ln(-x) & x < -1 \end{cases}$$

H. Use the IVT to show that each equation has a solution on the given interval.

$$1. \tan(\cos(x)) = \frac{1}{x^2+1}, [0, 2]$$

$$2. x^2 = e^x + 4, [-3, 0]$$

$$3. \ln(x^2 - 1) = \csc(x), [6, 8]$$